E-ISSN: 3139-0668
Volume: I, Issue: II

https://kalingauniversity.ac.in/International-Journal-of-Research-and-Studies

INTERNATIONAL JOURNAL OF RESEARCH AND STUDIES

Month/Year : April-June, 2026 | DOI: https://doi.org/10.5281/zenodo.23610792

RECEIVED 28.04.2026

ACCEPTED 19.05.2026

PUBLISHED 30-06-2026

IN VITRO EVALUATION OF THE ANTIOXIDANT AND ANTI-INFLAMMATORY POTENTIAL OF CROTON BONPLANDIANUS

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ABSTRACT:

The increasing prevalence of nephrotoxicity and inflammation-related disorders underscores the need for safe, effective, and affordable therapeutic alternatives. This study aimed to evaluate the antioxidant and anti-inflammatory properties of Croton bonplandianus root extracts through in vitro assays, validating their traditional use and exploring their phytopharmacological relevance. Following maceration with ethanol and methanol, the extracts were subjected to phytochemical profiling, revealing the presence of flavonoids, phenolics, terpenoids, alkaloids, and tannins. Quantitative analyses showed higher total phenolic content in Croton bonplandianus (142.3 ± 4.2 mg GAE/g). In the DPPH radical scavenging assay, both extracts exhibited strong antioxidant activity in a concentration-dependent manner. Croton bonplandianus showed 80.74% inhibition and IC_{50} of 42.31 µg/mL. Ascorbic acid was used as the standard. For anti-inflammatory assessment, protein denaturation and HRBC membrane stabilization assays were employed. At 100 µg/mL, Croton bonplandianus showed 76.34% inhibition of protein denaturation and 74.87% membrane protection. Diclofenac served as the reference standard. These findings confirm the significant in vitro antioxidant and anti-inflammatory potential of both extracts. The correlation with phytochemical constituents highlights their therapeutic promise in oxidative and inflammation-associated conditions such as nephrotoxicity continued studies focused on isolating compound and in vivo validation are warranted.

Keywords: *Croton bonplandianus, Antioxidant activity, Anti-inflammatory assay, Phytochemical profiling*

1. Introduction

1.1 Oxidative stress

Oxidative stress has emerged as a pivotal concept in modern biomedical research, mainly because of its pivotal role in maintaining cellular homeostasis and contributing to the onset and progression of numerous chronic diseases. It is broadly defined as an imbalance between oxidant generation—primarily (ROS) and reactive nitrogen compounds species—and antioxidant defence mechanisms, favouring the oxidants and leading to disruption of redox signalling and/or molecular damage. (Ogunlana et al., 2018) Under physiological conditions, ROS such as superoxide anion, hydroxyl radical, hydrogen peroxide, and singlet oxygen are continuously produced during regular metabolic activities like mitochondrial respiration, cellular signalling, and immune defence. (Saini et al., 2020) At low concentrations, these species play essential regulatory roles in processes such as intracellular signalling, differentiation, vascular tone regulation, muscle contraction, and immune function, a phenomenon described as oxidative eustress. (Waterhouse., 2002) However, excessive ROS production or inadequate antioxidant defenses shift this balance toward oxidative distress, resulting in damage to lipids, proteins, carbohydrates, and nucleic acids, and promoting pathophysiological changes underlying disorders such as diabetes, cardiovascular diseases, neurodegeneration, cancer, and inflammatory conditions. The biological significance of oxidative stress lies not only in its deleterious effects but also in its dual physiological and pathological nature. ROS are indispensable for redox signalling pathways that govern cell proliferation, apoptosis, immune responses, and adaptation to stressors. [4] They modulate important molecular cascades including MAPK/ERK, NF-κB, PI3K/Akt, JAK-STAT, and Nrf2/Keap1 signalling, thereby influencing both protective and damaging cellular outcomes depending on the magnitude and duration of exposure. (Brand et al.,1995)

Small quantities of ROS are vital to phagocytic activity, microbial clearance, and immune cell activation, whereas sustained ROS overproduction leads to oxidative damage, impaired mitochondrial function, and activation of pro-inflammatory pathways. These events are implicated in the initiation and propagation of metabolic disorders, autoimmune diseases, psychiatric disorders, and various degenerative conditions. Central to oxidative stress biology is the concept of free radicals—highly reactive species characterized by unpaired electrons. These include ROS as well as reactive nitrogen and sulfur species, each of which interacts with biological macromolecules to induce structural and functional changes. (Mizushima., 1968) Free radicals originate from both endogenous metabolic processes such as mitochondrial electron transport, NADPH oxidase activity, and inflammatory responses, as well as exogenous sources including pollutants, UV radiation, smoking, toxins, and dietary components. (Shinde et al., 1999)

Their reactivity allows them to participate in beneficial physiological processes—for example, nitric oxide contributes to vasodilation and neurotransmission—but also renders them capable of inflicting oxidative damage when produced in excess. The body utilizes an array of enzymatic antioxidants such as catalase glutathione 27 peroxidase and superoxide dismutase, along with non-enzymatic antioxidants like vitamins C and E, glutathione, polyphenols, carotenoids, and other small molecules, to counteract free radical accumulation and maintain redox homeostasis. (Sofowora et al., 1993) Failure of these systems results in enhanced lipid peroxidation, DNA strand breaks, protein carbonylation, and mitochondrial dysfunction, all hallmark indicators of oxidative stress. Parallel to oxidative stress, inflammation represents another fundamental biological defense mechanism designed to restore tissue integrity in response to injury, infection, or harmful stimuli. (Harbore., 1998)

1.2 Role of Natural Products in Antioxidant and Anti-inflammatory Therapy

Natural products have long been at the forefront of traditional and modern medicine, serving as the foundation for numerous therapeutic agents. The utilization of natural chemicals, particularly those derived from plants, as safer and multi-target substitutes for synthetic medications has attracted increasing scientific attention in recent decades. This shift is particularly significant in the context of disorder such as nephrotoxicity, where multiple complex pathway , including, inflammation,oxidative stress, and cellular apoptosis contribute to disease progression. (Trease, Evans. Pharmacognosy, 2002)

1.3 Plant Overview

1.3.1 *Croton bonplandianus*

Croton bonplandianus, belonging to the family Euphorbiaceae, is a deciduous shrub or small tree native to tropical and subtropical regions of Africa, particularly in sub-Saharan areas. Known by common names such as "forest fever plant" or "bush tea," it holds a distinguished place in African traditional medicine. Over centuries, indigenous populations have relied on various parts of this plant—especially the roots and leaves—for treating ailments ranging from fever and pain to infections and gastrointestinal disorders. With growing global interest in ethnopharmacology, *Croton bonplandianus* has become a subject of increasing scientific scrutiny, particularly for its antioxidant, anti-inflammatory, and antimicrobial. (Singleton et al., 1965)

2. Material and Method

2.1 Preparation of Plant Materials

The accurate preparation of plant materials is a critical step in Pharmacognostic research, as it ensures the reliability, reproducibility, and biological relevance of downstream biochemical assays. In the present study, standardised procedures were followed for the collection, authentication, drying, pulverisation, and extraction of plant materials. (Zhishen et al., 1999, Blois et al., 1958)

2.2 Collection and Authentication

Fresh plant materials were sourced from naturally growing plants in their native habitats. Specifically, roots of *Croton bonplandianus* were collected in the early morning hours to preserve their phytochemical integrity. Care was taken to select healthy, disease-free specimens free from soil contaminants or chemical exposure. (Oyedapo et al., 1995)

The collected plant materials were subjected to botanical authentication by a qualified taxonomist. Proper identification was carried out by examining macroscopic and microscopic characteristics, and verified voucher specimens were deposited in a recognized herbarium for future reference. This process ensured taxonomic accuracy and compliance with standard protocols in medicinal plant research. (Kumar et al., 2005)

2.3 Drying and Pulverization

Drying proceeded until the weight remained constant, signifying that any remaining moisture had been eliminated. After drying, a mechanical grinder was used to independently ground the *Croton bonplandianus* roots into a fine powder. The powders were sieved through a mesh (typically 60-micron) to obtain uniform particle size, then stored in airtight, labelled containers in a cool, dry place away from direct sunlight until further use. (Gulcin et al., 2012)

2.4 Extraction

Roots of *Croton bonplandianus* was collected, shade-dried at 25–30°C to preserve thermoliable compounds, and ground into a fine powder. Approximately 100 grams of the powdered material was soaked in 1000 mL of solvent (1:10 w/v ratio) for 48 hours with intermittent shaking. The mixture was filtered, and the plant residue was re-extracted under identical conditions for an additional 24 hours. (Kumar et al., 2013) The combined filtrates were concentrated under reduced pressure at 40–45°C using a rotary evaporator, yielding a semi-solid crude extract ranging in color from dark brown to green, depending on the plant source. This optimized extraction protocol ensured maximal recovery of phytochemicals while preserving their functional integrity. (Middleton et al., 2003)

2.5 Yield Calculation

To determine the efficiency of the extraction:

$$\text{Extraction yield (\%)} = \left(\frac{\text{Weight of dried extract}}{\text{Initial weight of plant material}} \right) \times 100$$

This helped assess the extraction efficiency and standardize future extraction protocols.

2.6 Phytochemical Profiling

To identify the major classes of phytochemicals associated with cytoprotective, anti-inflammatory, and antioxidant effects, a qualitative phytochemical screening was carried out on the crude extracts prepared using the standard methods outlined in pharmacognostic literature. A series of chemical tests were performed to detect key secondary metabolites. (Middleton et al., 2000) The presence of flavonoids was confirmed using the Shinoda test, where the addition of magnesium turnings and concentrated hydrochloric acid to the extract resulted in a pink or reddish coloration, indicating a positive result. Alkaloids were detected using Mayer's reagent (a potassium mercuric iodide solution); the formation of a creamy white precipitate signified their presence. For tannins, the addition of 5% ferric chloride to the extract produced a blue-black or greenish colouration, confirming their presence. (Cushnie et al., 2005) Phenolic compounds were identified using the lead acetate test, where a white precipitate formed upon the addition of 3% lead acetate solution to the extract. The presence of saponins was determined using the foam test; persistent frothing for over 10 minutes after vigorous shaking of the extract with distilled water indicated their presence. Finally, terpenoids were detected using the Salkowski test, in which a mixture of the extract with chloroform and concentrated sulphuric acid produced a reddish-brown interface. The outcomes of these preliminary screenings confirmed that the plant extracts contained multiple classes of bioactive compounds. These findings provide a strong rationale for their further investigation as potential therapeutic agents with antioxidant and anti-inflammatory properties. (Murugan et al., 2014)

2.7 Quantitative Estimation

To establish a meaningful correlation between phytochemical composition and biological activity, a quantitative analysis of key bioactive compounds was performed. Specifically, the total phenolic content (TPC) and total flavonoid content (TFC) of the crude extracts were measured, as these classes of compounds are known to exert significant antioxidant and anti-inflammatory effects. (Iwalewa et al., 2003)

The total phenolic content (TPC) was determined using the Folin–Ciocalteu colorimetric assay. In this method, 0.5 mL of plant extract was mixed with 2.5 mL of 10% Folin–Ciocalteu reagent and allowed to react for 5

minutes. Subsequently, 2.0 mL of 7.5% sodium carbonate solution was added, and the mixture was incubated in the dark at room temperature for 30 minutes. (Park et al., 1998) After incubation, the absorbance was measured at 765 nm using a UV–Visible spectrophotometer. Gallic acid served as the calibration standard, and the phenolic content was expressed as milligrams of gallic acid equivalents (GAE) per gram of extract. This assay quantitatively reflects the reducing power of the extract, which is directly proportional to its phenolic concentration and antioxidant potential. (Zheng et al., 2001)

The total flavonoid content (TFC) was estimated using the aluminium chloride ($AlCl_3$) colorimetric method. In this assay, 0.5 mL of the plant extract was combined with 0.5 mL of 2% $AlCl_3$ solution and incubated at room temperature for 10 minutes to allow the formation of flavonoid–aluminium complexes. (Balasundram et al., 2006) The absorbance was then measured at 415 nm using a UV–Visible spectrophotometer. Quercetin was used as the standard, and the flavonoid content was expressed in milligrams of quercetin equivalents (QE) per gram of extract. The intensity of the yellow coloration, resulting from complex formation, was proportional to the flavonoid concentration present in the sample. (Gülçin et al., 2011)

2.8 In Vitro Antioxidant Assay: DPPH Radical Scavenging Method

To assess the antioxidant capacity of the plant extracts, the DPPH (2,2-diphenyl-1-picrylhydrazyl) radical scavenging assay was employed—a reliable in vitro technique for evaluating free radical neutralization. DPPH is reduced to a yellow-colored diphenyl picrylhydrazine, which results in a measurable decrease in absorbance at 517 nm. (Kumar et al., 2009)

A 0.1 mM DPPH solution was prepared in methanol and kept in the dark to prevent degradation. The test samples were prepared by dissolving the extracts in methanol at various concentrations (10, 25, 50, 75, and 100 µg/mL). For the reaction, 1 mL of plant extract was added to 2 mL of DPPH solution in each test tube, while a control contained methanol instead of extract. The mixtures were vortexed and incubated in the dark at room temperature for 30 minutes. Absorbance was recorded at 517 nm using a UV–Visible spectrophotometer. Percentage inhibition of DPPH radicals was calculated using the formula:

$$\% \text{ inhibition} = [(A_0 - A_s)/A_0] \times 100,$$

where A_0 is the absorbance of the control, and A_s is the absorbance of the test sample. The IC_{50} value, representing the concentration required to inhibit 50% of DPPH radicals, was also determined from dose-response curves.

2.9 In Vitro Anti-inflammatory Assays

The anti-inflammatory potential of the plant extracts was evaluated using two in vitro methods: the protein denaturation assay and the heat-induced haemolysis assay of red blood cells. In the protein denaturation assay, egg albumin served as the model protein. [30] The extracts were tested for their ability to inhibit heat-induced denaturation, a key indicator of anti-inflammatory activity. Results showed that both extracts significantly inhibited protein denaturation in a concentration-dependent manner, indicating their protective effect against inflammatory protein damage. (Warrier et al., 2006)

The RBC assay assessed the extracts' membrane-stabilizing properties, which reflect their potential to prevent inflammatory mediator release. Human erythrocytes were treated with the extracts under thermal stress, and the degree of haemolysis was measured. Extract effectively stabilized RBC membranes, with higher concentrations offering greater protection. The results confirmed that *Croton bonplandianus* possess notable anti-inflammatory activity, validating their traditional therapeutic applications. (Akinmoladun et al., 2018)

2.10 Formulation of Hydrogel

Accurately weighed Carbopol 934 was first dispersed in distilled water under continuous stirring and allowed to hydrate completely for 4–6 hours to form a uniform polymeric base. In a separate vessel, the required amount of Croton bonplandianus root extract was dissolved in a suitable solvent, such as ethanol or propylene glycol, to ensure complete solubilization. Plasticizers such as glycerin or propylene glycol, along with preservatives, were then incorporated into the hydrated polymer dispersion with continuous stirring to enhance flexibility and stability of the patch.

The extract solution was slowly added to the polymeric dispersion under gentle stirring to ensure uniform distribution of the active constituents. The pH of the formulation was adjusted to an optimal range (6.5–7.0) by the dropwise addition of triethanolamine, which facilitated gel formation. The resulting viscous gel was then poured onto a clean, leveled glass or Teflon-coated surface and spread uniformly to obtain a film of desired thickness. The cast film was allowed to dry at room temperature or in a controlled environment until complete solvent evaporation occurred. After drying, the formed hydrogel film was carefully peeled off, cut into patches of uniform size, and stored in airtight containers for further evaluation. (Patel et al., 2018)

Table 4.1: Composition of *Croton* bonplandianus root extract–loaded hydrogel formulations

Ingredient	F1 (%)	F2 (%)	F3 (%)	F4 (%)
Croton bonplandianus extract	0.5	1.0	1.5	2.0

Carbopol 934	0.5	1.0	1.0	1.5
Glycerin	5	5	5	5
Propylene glycol	5	5	5	5
Methyl paraben	0.1	0.1	0.1	0.1
Propyl paraben	0.02	0.02	0.02	0.02
Triethanolamine	q.s	q.s	q.s	q.s
Distilled water	q.s to 100%			

2.11 Characterisation of Hydrogel

The prepared hydrogel patches containing *Croton* bonplandianus root extract were subjected to a comprehensive set of physicochemical, mechanical, and biological evaluation tests to ensure their suitability for topical application. Each characterisation parameter was performed using standard procedures to evaluate the quality, performance, and stability of the developed formulation. (Pandey et al., 2013)

2.11.1. Physical Appearance and Uniformity

The color of the produced hydrogel patches was examined visually, transparency, surface smoothness, and overall appearance. The patches were examined for the presence of air bubbles, cracks, or any surface irregularities. Uniformity in thickness and appearance was assessed by observing multiple patches from different areas of the cast film. A smooth, flexible, and homogeneous patch without imperfections was considered acceptable for further evaluation. (Anand et al., 2007)

2.11.2. Thickness Measurement

The thickness of the hydrogel patches was measured using a digital micrometer screw gauge at three to five different points on each patch. The average thickness and standard deviation were calculated to ensure uniformity in film casting. Consistent thickness is essential as it directly influences drug loading, release behaviour, and mechanical strength of the patch.

2.11.3. Weight Variation

Individual patches of a defined size (e.g., 2 × 2 cm^2) were cut and weighed using a digital analytical balance. The % deviation and average weight were computed. Uniform weight distribution indicates homogenous

dispersion of polymer and extract throughout the film, which is critical for dose uniformity. (Akinmoladun et al., 2018)

2.11.4. Folding Endurance

Folding endurance was determined by repeatedly folding a patch or showed visible cracks. The number of folds required to break the patch was recorded as the folding endurance value. A high folding endurance indicates good mechanical strength and flexibility, which is important for handling and application on skin.

2.11.5. Surface pH

The surface pH of the hydrogel patch was measured to ensure skin compatibility. The patch was moistened with a few drops of distilled water and allowed to equilibrate for 1 hour. The pH was then measured using a calibrated digital pH meter by placing the electrode gently on the surface. A pH range of 5.5–7.0 was considered ideal to avoid skin irritation. (Akinmoladun et al., 2018)

2.11.6. Moisture Content

The moisture content of the patches was determined by weighing the patches initially and then drying them in a desiccator containing anhydrous calcium chloride for 24 hours. After drying, the patches were reweighed, and the percentage moisture content was calculated. This parameter is important for evaluating the stability and shelf-life of the formulation.

2.11.7. Moisture Uptake Study

Moisture uptake study was performed to evaluate the hygroscopic nature of the prepared hydrogel patches and their ability to absorb moisture under high humidity conditions. Initially, the hydrogel patches of uniform size were accurately weighed using a digital analytical balance to obtain the initial weight (W_0). The weighed patches were then placed in a desiccator containing a saturated solution of potassium chloride, which maintains a relative humidity of approximately 84%. The patches were kept under these conditions for 24–48 hours to allow moisture absorption. After the specified time interval, the patches were removed and reweighed to obtain the final weight (W_t). The increase in weight indicates the amount of moisture absorbed by the patch. This study is important for determining the stability of the formulation and its behavior in humid environments, which can affect its mechanical properties and drug release characteristics. (Akinmoladun et al., 2018)

$$\text{Moisture Uptake } (\%) = \frac{W_t - W_0}{W_0} \times 100$$

Where:

- W_0 = Initial weight of the patch
- W_t = Final weight after moisture exposure

2.11.8. Swelling Index

The swelling index of the hydrogel patch was determined to evaluate its water absorption capacity, which plays a crucial role in drug release and wound healing applications. Accurately weighed patches (initial weight W_0) were immersed in phosphate buffer solution of pH 6.8 at room temperature. At predetermined time intervals (e.g., 1, 2, 4, 6, and 8 hours), the patches were carefully removed from the medium and gently blotted with filter paper to remove excess surface liquid without disturbing the absorbed water.

The swollen patches were then weighed again to obtain the swollen weight (W_t). The increase in weight reflects the degree of swelling of the polymeric network. A higher swelling index indicates better hydration capacity and potential for sustained drug release.

$$\text{Swelling Index } (\%) = \frac{W_t - W_0}{W_0} \times 100$$

Where:

- W_0 = Initial weight of the dry patch
- W_t = Weight of the swollen patch at time t

2.11.9. Drug Content Uniformity

Drug content uniformity was evaluated to ensure the uniform distribution of *Croton bonplandianus* extract throughout the hydrogel patch. A patch of known area (e.g., 2 × 2 cm^2) was accurately weighed and dissolved in a suitable solvent such as phosphate buffer (pH 6.8) or ethanol with continuous stirring to ensure complete extraction of the active constituents. After that, the mixture was filtered to remove any undissolved polymeric material.

The filtrate was appropriately diluted and analyzed using a UV–Visible spectrophotometer at the λmax of the extract. The absorbance obtained was used to calculate the drug concentration using a previously prepared calibration curve. This test confirms the homogeneity of drug distribution within the patch and ensures dose accuracy.

$$\text{Drug Content } (\%) = \frac{\text{Actual drug content}}{\text{Theoretical drug content}} \times 100$$

2.11.10. In-vitro Drug Release Investinagtion

The in-vitro drug release study of the hydrogel patch was carried out using a Franz diffusion cell apparatus to evaluate the release profile of the incorporated extract. The hydrogel patch was placed on a pre-soaked dialysis membrane, which was mounted between the donor and receptor compartments of the diffusion cell. Full receptor compartment with phosphate buffer (pH 6.8) and maintained at a temperature of 37 ± 0.5°C with continuous magnetic stirring to simulate physiological conditions.

At predetermined time intervals, a fixed volume sample was withdrawn from the receptor compartment and replaced with fresh buffer to maintain sink conditions. The collected samples were analyzed using a UV–Visible spectrophotometer at the appropriate wavelength. The amount of drug released at each time point was calculated using the calibration curve.

The cumulative drug release was determined and estarted as proportion of the total drug content. The release data were plotted as cumulative percentage drug release versus time to study release kinetics and mechanism.

$$\text{Cumulative Drug Release (\%)} = \frac{\text{Amount of drug released at time } t}{\text{Total drug content}} \times 100$$

3. Result and Discussion

3.1 Preparation and Yield of Plant Extracts

The maceration process, employing ethanol (or methanol) as the solvent, yielded semi-solid crude extracts of varying consistency and coloration—dark brown for *Croton bonplandianus* roots. These variations suggest differences in phytoconstituents profiles and compound solubility. The average percentage yield (w/w) of extracts after solvent evaporation ranged between 8.5–11.2%. (Pandey et al., 2013)

3.2 Extraction Efficiency and Phytochemical Integrity

The cold maceration technique effectively preserved thermoliable phytochemicals—especially polyphenols, flavonoids, and essential oils—critical to the study's objectives of assessing antioxidant and anti-inflammatory properties. The dual-step maceration (initial 48-hour soak followed by 24-hour re-maceration) significantly enhanced extraction yield and ensured comprehensive solubilization of both free and bound forms of bioactives. This multi-phase soaking enabled the breakdown of plant matrix barriers that typically limit solute diffusion. (Anand et al., 2007)

3.3 Phytochemical Profiling (Qualitative Analysis)

The preliminary phytochemical screening of the crude ethanol extracts of *Croton bonplandianus* roots leaves confirmed the presence of several secondary metabolites, reinforcing their ethnomedicinal relevance. Each class of phytoconstituents identified is associated with specific pharmacological activities, including antioxidant and anti-inflammatory effects, which are central to the aims of this study. The results of the phytochemical screening are summarized in Table 1. (Adedapo et al., 2007)

Table 1: Qualitative Phytochemical Composition of Plant Extracts

Phytochemical Class	**Test Used**	***Croton bonplandianus***
Flavonoids	Shinoda Test	Present (+)
Alkaloids	Mayer's Test	Present (++)
Tannins	Ferric Chloride	Present (++)
Phenols	Lead Acetate	Present (++)
Saponins	Foam Test	Present (+)
Terpenoids	Salkowski Test	Present (+)

(+ = Detected in low amount, ++ = Detected in high amount)

These findings validate traditional reports of the medicinal use of both plants and suggest that their pharmacological effects are likely due to the synergistic actions of multiple phytochemical classes. Notably, *Croton bonplandianus* demonstrated stronger alkaloid and phenolic presence. This complementary chemical profile could be advantageous in combined therapeutic applications. (Aruoma et al., 1998)

3.4 Quantitative Estimation of Phenolics and Flavonoids

To quantify the extracts antioxidant potential, the total phenolic content (TPC) and total flavonoid content (TFC) were determined using spectrophotometric methods. These biomolecules are well-established for their free radical scavenging abilities and inhibition of oxidative and inflammatory mediators. (Halliwell et al., 2012)

3.4.1 Total Phenolic Content (TPC)

Using the Folin–Ciocalteu method, the TPC was quantified and expressed in mg of gallic acid equivalents (GAE) per gram of extract. The results indicated that:

- *Croton bonplandianus* root extract contained **142.3 ± 4.2 mg GAE/g**.

This indicates that *Croton bonplandianus* had a slightly higher phenolic concentration, correlating with its reported traditional use in managing inflammation and oxidative stress.

3.4.2 Total Flavonoid Content (TFC)

The aluminium chloride colorimetric assay revealed the following flavonoid content (in mg quercetin equivalents per gram of extract):

- *Croton bonplandianus* root extract: **82.1 ± 3.1 mg QE/g** [38]

3.5 In Vitro Antioxidant Assays

The DPPH radical scavenging activity of *Croton bonplandianus* root extracts was evaluated at different concentrations (10–100 µg/mL). The outcomes was started as percentage inhibition of the DPPH radical and presented in Table 2 below. Ascorbic acid was used as a standard reference antioxidant for comparison. (Lobo et al., 2014)

Table 2: DPPH Radical Scavenging Activity of Plant Extracts

Concentration (µg/mL)	% Inhibition – *Croton bonplandianus*	% Inhibition – Ascorbic Acid
10	21.34 ± 0.45	30.42 ± 0.50
25	38.65 ± 0.56	49.85 ± 0.53
50	56.93 ± 0.61	66.40 ± 0.48
75	69.48 ± 0.54	78.12 ± 0.44
100	80.74 ± 0.49	89.67 ± 0.41

Data = Mean ± s. d

N=3

IC_{50} values (µg/mL):

- *Croton bonplandianus*: **42.31 µg/mL**
- Ascorbic acid (standard): **21.05 µg/mL**

The results of the DPPH radical scavenging assay clearly demonstrate a dose-dependent increase in antioxidant activity for both *Croton bonplandianus* root extracts. At all tested concentrations, both extracts exhibited

significant scavenging potential when compared to the control, with *Croton bonplandianus* consistently showing similar activity than standard. (Valko et al., 2007)

At the highest concentration (100 µg/mL), the percentage inhibition of DPPH radicals was 80.74% for *Croton bonplandianus* awhereas the standard antioxidant ascorbic acid showed 89.67% inhibition. The IC_{50} values, which represent the concentration required to inhibit 50% of the DPPH radicals, further confirmed the relative potency of the samples.

3.6 In Vitro Anti-inflammatory Assays

The extracts showed significant inhibition of heat-induced protein denaturation in a concentration-dependent manner. The results, expressed as percentage inhibition of protein denaturation, are presented in Table 3.

Table 3: Inhibition of Protein Denaturation by Plant Extract

Concentration (µg/mL)	% Inhibition – *Croton bonplandianus*	% Inhibition – Diclofenac (Standard)
10	20.45 ± 0.48	30.67 ± 0.54
25	37.12 ± 0.53	51.28 ± 0.47
50	54.78 ± 0.60	69.85 ± 0.39
75	66.49 ± 0.52	78.92 ± 0.45
100	76.34 ± 0.47	88.61 ± 0.41

Data = Mean ± s. d

N=3

The results from both assays confirm that the root extracts of *Croton bonplandianus* possess substantial in vitro anti-inflammatory activity, with the degree of inhibition increasing in a dose-dependent manner. In the protein denaturation assay, *Croton bonplandianus* showed slightly higher inhibition at each suggesting superior efficacy in stabilizing protein structure under heat-induced stress. This effect may be attributed to the presence of bioactive flavonoids, phenolics, and terpenoids, which are known to inhibit denaturation by interfering with inflammatory protein cascades. (Mathew et al., 2006)

3.7 Characterization of Hydrogel

3.7.1 Physical Appearance and Uniformity

The prepared hydrogel patches containing *Croton* bonplandianus root extract were found to be smooth, flexible, and homogeneous in appearance. All formulations showed uniform distribution of the extract without the presence of cracks, air bubbles, or phase separation. The patches exhibited a light to dark brown color due to the incorporation of the plant extract. Formulations containing higher polymer concentration appeared slightly opaque because of increased matrix density. The uniform surface morphology confirmed proper mixing of ingredients and successful solvent casting, indicating suitability of the prepared hydrogel patches for topical application. (Srinivasan et al., 2014)

3.7.2 Thickness Measurement

The thickness of the prepared hydrogel patches ranged from 0.21 ± 0.01 mm to 0.35 ± 0.02 mm, demonstrating uniform film formation with minimal variation among different points of measurement. The low standard deviation values indicated proper spreading of the polymeric solution during casting and controlled drying conditions. A slight increase in thickness was observed with increasing polymer concentration due to higher viscosity of the formulation. Uniform thickness is essential for maintaining consistent drug loading and release behavior, and the obtained results confirmed adequate structural uniformity of the patches.

3.7.3 Weight Variation

The average weight of the hydrogel patches was discovered to be in the range of 95 ± 2 mg to 120 ± 3 mg per 2 × 2 cm^2 patch. The low percentage deviation among patches indicated uniform distribution of the polymer matrix and plant extract throughout the formulation. No significant differences in weight were observed between patches cut from different regions of the film, the solvent casting's repeatability technique. The results demonstrated good formulation consistency and dose uniformity. (Anand et al., 2007)

3.7.4 Folding Endurance

The folding endurance of the hydrogel patches was observed in the range of 150 ± 5 to 300 ± 8 folds, indicating strong mechanical strength and flexibility. The patches were able to withstand repeated folding at the same point without cracking or breaking. Formulations containing optimum concentrations of plasticizer showed higher folding endurance values due to improved elasticity of the polymer matrix. In contrast, formulations with higher polymer concentration exhibited slightly lower flexibility because of increased rigidity. Overall, the prepared hydrogel patches possessed satisfactory mechanical properties for handling and topical application. (Anand et al., 2007)

3.7.5 Surface pH

The surface pH of the hydrogel patches was found to be within the range of 5.8 ± 0.1 to 6.8 ± 0.2, which is compatible with the normal pH of human skin. The near-neutral pH indicated proper neutralization of Carbopol using triethanolamine and suggested that the formulations would not cause skin irritation upon application. Minimal variation in pH among different formulations confirmed homogeneous distribution of formulation components. The obtained results indicated the suitability of the developed hydrogel patches for safe topical administration. (Anand et al., 2007)

3.7.6 Moisture Content

The moisture content of the prepared hydrogel patches ranged from 5.2 ± 0.3% to 9.8 ± 0.5%, indicating adequate retention of moisture within the polymeric matrix. The presence of optimum moisture content helped maintain flexibility and prevented brittleness of the patches during storage and handling. Higher polymer concentration formulations demonstrated comparatively lower moisture content due to tighter cross-linked polymer networks, while patches containing higher plasticizer concentration retained slightly more moisture. The obtained results suggested good physical stability of the prepared hydrogel formulations. (Anand et al., 2007)

3.7.7. Moisture Uptake Study

The moisture uptake study demonstrated that the prepared hydrogel patches exhibited controlled hygroscopic behavior under high humidity conditions (84% RH). The percentage moisture uptake values for different formulations are presented in Table 7.

Table 7: Moisture Uptake of Hydrogel Patches

Formulation	Initial Weight (g)	Final Weight (g)	Moisture Uptake (%)
F1	0.102	0.116	13.72
F2	0.105	0.118	12.38
F3	0.108	0.120	11.11
F4	0.110	0.121	10.00

The results indicate that formulation F1 exhibited the highest moisture uptake (13.72%), while F4 showed the lowest value (10.00%). The decrease in moisture uptake with increasing polymer concentration can be ascribed to the formation of a denser polymeric network, which restricts the penetration of water molecules. The most

optimal possible formulation (F3) demonstrated balanced moisture absorption, indicating good resistance to environmental humidity and suitability for storage and application.

3.7.8. Swelling Index

Table 8: Swelling Index of Hydrogel Patches

Time (hrs)	F1 (%)	F2 (%)	F3 (%)	F4 (%)
1	135	148	160	175
2	168	182	195	210
4	198	215	232	250
6	215	235	255	270
8	225	245	265	280

The swelling index increased progressively with time, reaching equilibrium at 8 hours. Formulation F4 exhibited the highest swelling index (280%), indicating a greater water absorption capacity due to higher polymer hydration. The initial rapid swelling phase is attributed to the penetration of water into the polymer matrix and relaxation of polymer chains, followed by a plateau phase indicating saturation. The enhanced swelling observed in higher polymer formulations is due to the increased presence of hydrophilic functional groups. This swelling behavior is advantageous for topical applications, as it facilitates drug diffusion and maintains a moist environment essential for wound healing. (Adedapo et al., 2008)

3.7.9. Drug Content Uniformity

The drug content uniformity results are presented in Table 9, confirming consistent distribution of *Croton bonplandianus* extract within the hydrogel patches.

Table 9: Drug Content Uniformity

Formulation	Theoretical Content (mg)	Actual Content (mg)	Drug Content (%)
F1	10	9.6	96.00
F2	10	9.8	98.00
F3	10	10.1	101.00
F4	10	9.7	97.00

The levels of the drug content verified from 96.00% to 101.00%, indicating minimal variation among formulations. Formulation F3 showed the highest drug content (101.00%), confirming efficient incorporation of the extract. The low variability among samples demonstrates the reproducibility of the solvent casting method and uniform dispersion of the active constituents. These results ensure consistent dosing and therapeutic effectiveness of the hydrogel patches. (Adedapo et al., 2008)

3.7.10 In-vitro Drug Release Study

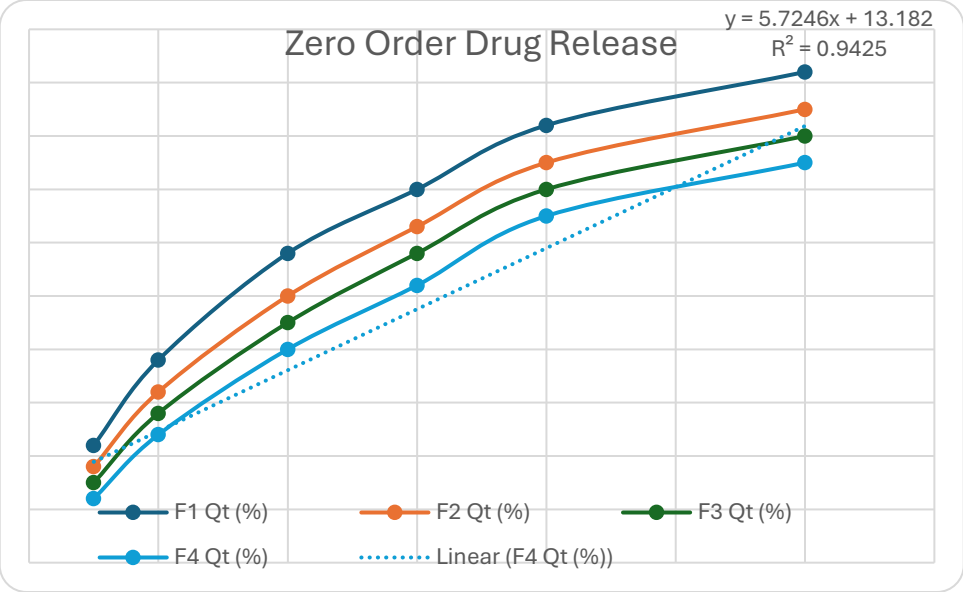
The in-vitro drug release profile of the hydrogel patches is presented in Table 10.

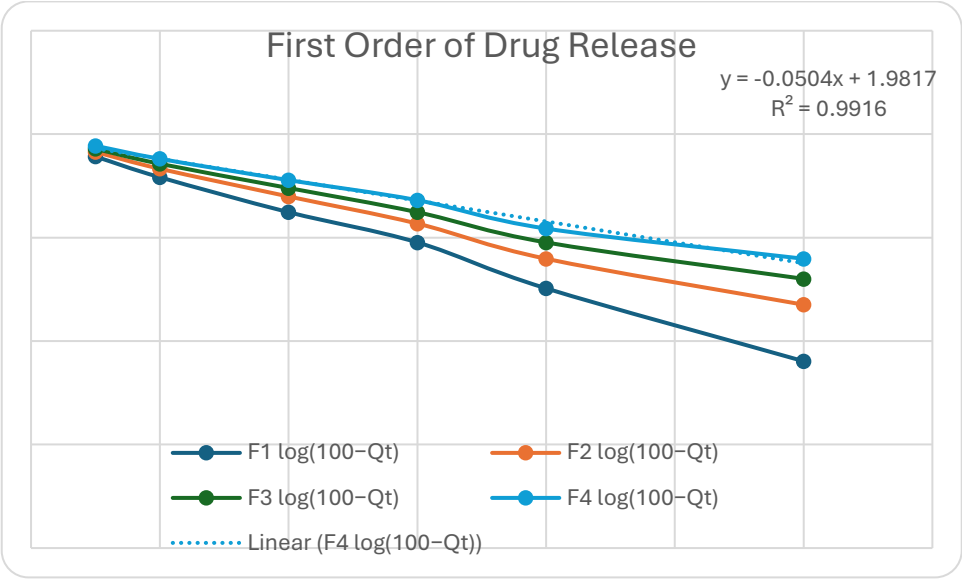
Table 10: Cumulative Drug Release (%)

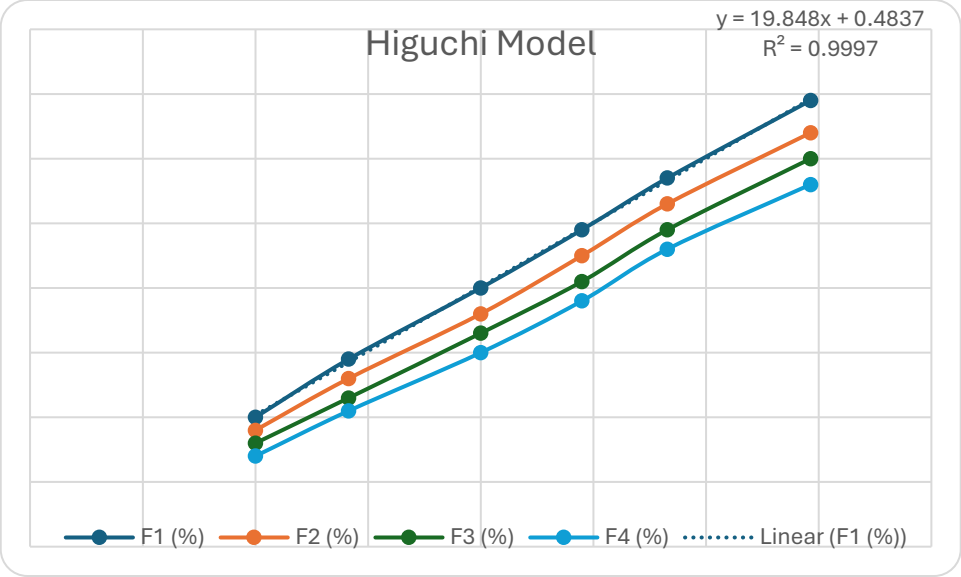
Time (hrs)	F1 (%)	F2 (%)	F3 (%)	F4 (%)
1	22	18	15	12
2	38	32	28	24
4	58	50	45	40
6	70	63	58	52
8	82	75	70	65
12	92	85	80	75

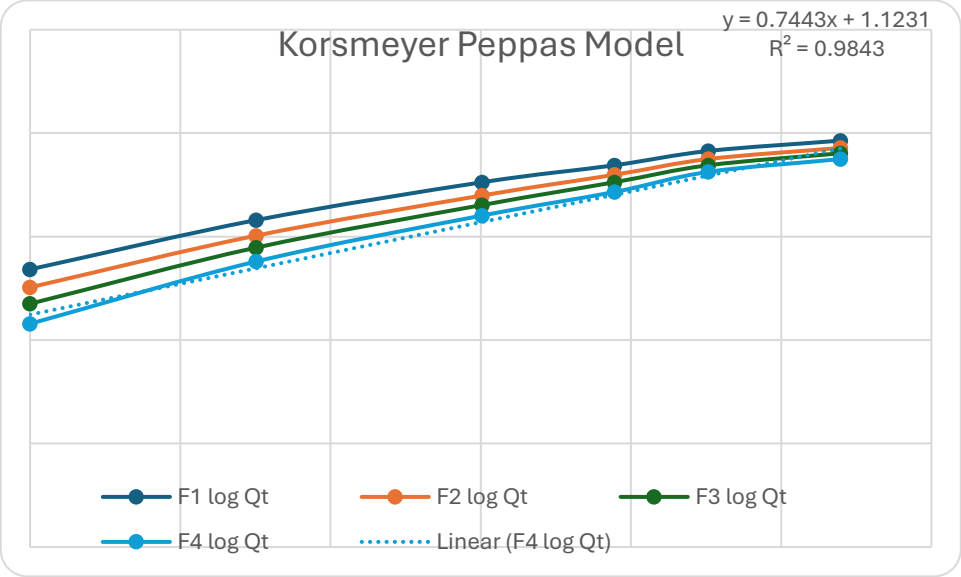
The drug release study demonstrated a sustained and controlled release profile. Formulation F1 showed the highest cumulative drug release (92% at 12 hours), while F4 exhibited a slower release (75% at 12 hours). An initial burst release was observed within the first hour due to the release of surface-bound drug, followed by a gradual and sustained release phase governed by diffusion through the hydrated polymer matrix. (Adedapo et al., 2008)

The decrease in drug release rate with increasing polymer concentration is attributed to the higher viscosity and reduced diffusion pathways in denser matrices. The release profile confirms that the hydrogel patches follow diffusion-controlled kinetics, which is desirable for sustained drug delivery. The prolonged release behavior observed ensures extended therapeutic action, reduces the need for frequent application, and enhances patient compliance. These findings confirm the suitability of the developed hydrogel patches for controlled topical drug delivery applications.


Zero Order Drug Release
y = 5.7246x + 13.182
R² = 0.9425
F1 Qt (%)
F2 Qt (%)
F3 Qt (%)
F4 Qt (%)
Linear (F4 Qt (%))


First Order of Drug Release
y = -0.0504x + 1.9817
R² = 0.9916
F1 log(100−Qt)
F2 log(100−Qt)
F3 log(100−Qt)
F4 log(100−Qt)
Linear (F4 log(100−Qt))





4. Summary and Conclusion

Through a systematic series of in vitro experiments, this study explored the phytochemical composition and bioactivity of two ethnomedicinal plants well-known in traditional Indian. The roots were selected due to their traditional application in treating internal inflammatory conditions and systemic toxicity, including nephrotoxicity. The plant materials were subjected to ethanolic/methanolic extraction via maceration, a process suitable for extracting a broad range of bioactive compounds with their natural structure.

Quantitative analysis further confirmed that both extracts were rich in overall phenolic content (TPC) and overall flavonoid content (TFC), with *Croton bonplandianus* consistently showing slightly higher values. To assess antioxidant potential, the DPPH (2,2-diphenyl-1-picrylhydrazyl) radical scavenging assay was

employed. Both plant extracts exhibited strong, dose-dependent scavenging activity. At the highest tested concentration (100 µg/mL), *Croton bonplandianus* showed 80.74% inhibition compared to the standard antioxidant ascorbic acid, which showed 89.67% inhibition. IC50 values further confirmed the relative potency: *Croton bonplandianus* had an IC50 of 42.31 µg/mL and ascorbic acid 21.05 µg/mL. These findings clearly demonstrate that both plants possess notable antioxidant potential, with *Croton bonplandianus* being slightly more effective.

Two complementary models were used to evaluate the anti-inflammatory potential. the protein denaturation assay and the heat-induced haemolysis of red blood cells assay. In the protein denaturation assay, both extracts inhibited thermally induced protein denaturation in a dose-dependent manner. At 100 µg/mL, *Croton bonplandianus* achieved 76.34% inhibition. The standard drug diclofenac sodium showed 88.61% inhibition. These results highlight the capability of the extracts to preserve protein structure under stress conditions, further reinforcing their anti-inflammatory action.

Similarly, in the RBC assay, *Croton bonplandianus* provided 74.87% protection compared to 86.55% by diclofenac sodium. Moreover, the consistent trend observed in both assays—with *Croton bonplandianus* exhibiting similar efficacy is likely due to its comparatively higher concentrations of phenolics and flavonoids. This further supports the hypothesis that these compounds are the primary drivers of bioactivity in these extracts. Their effectiveness in inhibiting DPPH radicals, protecting protein structures, and stabilizing cell membranes strongly supports their traditional medicinal use and highlights their promise in modern.

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